In Vitro and In Vivo Trypanocidal Efficacy of Synthesized Nitrofurantoin Analogs.

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African trypanosomes cause diseases in humans and livestock. Human African trypanosomiasis is caused by Trypanosoma brucei rhodesiense and T. b. gambiense. Animal trypanosomoses have major effects on livestock production and the economy in developing countries, with disease management depending mainly on chemotherapy. Moreover, only few drugs are available and these have adverse effects on patients, are costly, show poor accessibility, and parasites develop drug resistance to them. Therefore, novel trypanocidal drugs are urgently needed. Here, the effects of synthesized nitrofurantoin analogs were evaluated against six species/strains of animal and human trypanosomes, and the treatment efficacy of the selected compounds was assessed in vivo. Analogs 11 and 12, containing 11- and 12-carbon aliphatic chains, respectively, showed the highest trypanocidal activity ($IC50 < 0.34$ μM) and the lowest cytotoxicity ($IC50 > 246.02$ μM) in vitro. Structure-activity relationship analysis suggested that the trypanocidal activity and cytotoxicity were related to the number of carbons in the aliphatic chain and electronegativity. In vivo experiments, involving oral treatment with nitrofurantoin, showed partial efficacy, whereas the selected analogs showed no treatment efficacy. These results indicate that nitrofurantoin analogs with high hydrophilicity are required for in vivo assessment to determine if they are promising leads for developing trypanocidal drugs.

The Inhibition of Cysteine Proteases Rhodesain and TbCatB: A Valuable Approach to Treat Human African Trypanosomiasis.

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Human African Trypanosomiasis (HAT) is an endemic parasitic disease of sub-Saharan Africa, caused by two subspecies of protozoa belonging to Trypanosoma genus: T. brucei gambiense and T. brucei rhodesiense. In this context the inhibition of the papain-family cysteine proteases rhodesain and TbCatB has to be considered a promising strategy for HAT treatment. Rhodesain, the major cathepsin L-like cysteine protease of T. brucei rhodesiense, is a lysosomal protease essential for parasite survival. It is involved in parasite invasivity, allowing it to cross the blood-brain barrier (BBB) of the human host, causing the second lethal stage of the disease. Moreover, it plays an important role in immunoevasion, being involved in the turnover of variant surface glycoproteins of the T. brucei coat and in the degradation of immunoglobulins, avoiding a specific immune response by the host cells. On the other hand TbCatB, a cathepsin B-like cysteine protease, present in minor abundance in T. brucei, showed a key role in the degradation of host transferrin, which is necessary for iron acquisition by the parasite. In this review article we now discuss the most active peptide, peptidomimetic and non-peptide rhodesain and TbCatB inhibitors as valuable strategy to treat HAT, due also to the complementary role of the two T. brucei proteases.

[Human serum trypanosome lytic factor and serum resistance-associated protein of trypanosome].

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Human African trypanosomiasis is caused by the infection of Trypanosoma brucei gambiense or T.b.

rhodesiense, while another morphologically identical subspecies, T.b. brucei, and other closely related species, T. equiperdum and T. evansi, are considered not infectious to human. This is highly related to the trypanosome lytic factor (TLF) found in normal human serum (NHS) and the serum resistance-associated (SRA) protein of trypanosomes infectious to human. We reviewed the research progress in TLF and its role in trypanosome lysis as well as the mechanism of SRA against the TLF.

Human African trypanosomiasis: a review of non-endemic cases in the past 20 years.

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Human African trypanosomiasis (HAT) is caused by sub-species of the parasitic protozoan Trypanosoma brucei and is transmitted by tsetse flies, both of which are endemic only to sub-Saharan Africa. Several cases have been reported in non-endemic areas, such as North America and Europe, due to travelers, ex-patriots or military personnel returning from abroad or due to immigrants from endemic areas. In this paper, non-endemic cases reported over the past 20 years are reviewed; a total of 68 cases are reported, 19 cases of Trypanosoma brucei gambiense HAT and 49 cases of Trypanosoma brucei rhodesiense HAT. Patients ranged in age from 19 months to 72 years and all but two patients survived. Physicians in non-endemic areas should be aware of the signs and symptoms of this disease, as well as methods of diagnosis and treatment, especially as travel to HAT endemic areas increases. We recommend extension of the current surveillance systems such as TropNetEurop and maintaining and promotion of existing reference centers of diagnostics and expertise. Important contact information is also included, should physicians require assistance in diagnosing or treating HAT.

[Human African trypanosomiasis].

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Human African trypanosomiasis (HAT) is caused by infestation with a flagellate protozoan, the trypanosome which is inoculated by the bite of the tsetse fly Glossina. The particular ecological conditions of parasites and vectors are such that the disease is only found in the intertropical regions of Africa. Although there are many species of trypanosomes, only two, belonging to the brucei group are likely to lead to HAT. These two species are quite similar morphologically but have different pathogenicity. Trypanosoma brucei gambiense found in West and Central Africa leads to a chronic form of the disease or sleeping sickness. T. b. rhodesiense leads to a more virulent and acute condition, although for each species of trypanosome there are strains of different virulence, which account, at least in part, for the interindividual variability in the clinical course. Immediately after penetration into the human organism, the trypanosome multiplies at the point of inoculation, producing a local inflammatory reaction. It then invades the whole organism, and the central nervous system (CNS). The involvement of the CNS leads to an irreversible demyelinating process ending by death without treatment. Apart from the initial stages, it is not easy to determine the phase of the disease that the patient is presenting. The parasite can escape the host immune response by varying the surface glycoprotein coat. Variable surface glycoproteins (VSG) are strongly antigenic and lead to great antibody response with immune lysis. But, some heterologous antigenic variants can survive to repopulate blood and other tissues. This mechanism of antigenic variation is under parasite genetic control. The trypanosome can release numerous pathogenic substances which cause alterations in cytokine/prostaglandin network. A 41-46 kDa molecule termed trypanosome-released lymphocyte trigerring factor may selectively activate CD8+ T cells to produce interferon-gamma which then activates macrophages but also promotes parasite growth. Activated macrophages release tumor necrosis factor alpha and nitric oxide (NO) which are trypanostatic static and other cytokines and prostanglandins. These macrophage relased substances enhance the immunosuppression and alter the blood brain barrier (BBB). So, trypanosomes and inflammatory cells can invade the CNS leading to a progressive meningoencephalitis with typical perivascular cuffings which explain neurological disorders and neuroendocrine alterations. The inflammatory cells (lymphocytes, astrocytes, glial cells) produce cytokines, NO and other mediators and enhance the CNS immunopathological process. The peri-ventricular regions, the tuberoinfundibula and thalamic-hypothalamic regions, are particulary involved. These disturbances lead to a progressively complete disruption of the normal sleep-waking cycle. Antibodies anti-CNS components (galactocerebrosides, neurofilaments, tryptophane) are also described in sera and cerebrospinal fluid (CSF) of HAT patients. Their presence may be due to cross reactions with comon epitopes between host and trypanosomes which can lead to a self-propagating autoimmune reaction, which accounts for the marked demyelination found in the late stage of the disease. The diagnosis of CNS involvement in not easy to establish in the early neurological phase in the absence of neurological signs and in absence of great chnages in CSF. This is an important problem because it is the basis to apply existing available drugs. pentamidine and suramin are effective only in early stages of the disease when CNS is not invaded. Melarsoprol is effective in all-stages: this is the drug of choice when CNS is involved. Unfortunaley, melarsoprol is toxic and, in 5% of treated patients, this drug can lead to arsenical encephalopathy which is often fatal. In the continuing search for new antitrypanosonal drugs, biochemical peculiarities of the trypanosome are used as drug garget, especially glycolysis, trypanothione, sensibil

Recent Updates on Development of Drug Molecules for Human African Trypanosomiasis.

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Human African trypanosomiasis (HAT, better called as sleeping sickness), caused by two morphologically identicalprotozoan parasite Trypanosoma bruceiis transmitted by the bite of tsetse flies of Glossinagenus, mainly in the rural areas of the sub-Saharan Africa. HAT is one of the neglected tropical diseases and is characterized by sleep disturbance as the main symptom, hence is called as sleeping sickness. As it is epidemic in the poorest population of Africa, there is limited availability of safe and cost-effective tools for controlling the disease. Trypanosoma bruceigambiense causes sleeping sickness in Western and Central Africa, whereas Trypanosoma bruceirhodesiense is the reason for prevalence of sleeping sickness in Eastern and Southern Africa. For the treatment of sleeping sickness, only five drugs have been approved suramin, pentamidine, melarsoprol, eflornithine and nifurtimox. Various small molecules of diverse chemical nature have been synthesized for targeting HAT and many of them are in the clinical trialsincluding fexinidazole (phase I completed) and SCYX-7158 (advanced in phase I). The present work has been planned to review various types of small molecules developed in the last 10 years having potent antitrypanosoma activity likely to be beneficial in sleeping sickness along with different natural anti-HAT agents.

Trypanosoma brucei: Metabolomics for analysis of cellular metabolism and drug

discovery.

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Trypanosoma brucei is the causative agent of Human African Trypanosomiasis (also known as sleeping sickness), a disease causing serious neurological disorders and fatal if left untreated. Due to its lethal pathogenicity, a variety of treatments have been developed over the years, but which have some important limitations such as acute toxicity and parasite resistance. Metabolomics is an innovative tool used to better understand the parasite's cellular metabolism, and identify new potential targets, modes of action and resistance mechanisms. The metabolomic approach is mainly associated with robust analytical techniques, such as NMR and Mass Spectrometry. Applying these tools to the trypanosome parasite is, thus, useful for providing new insights into the sleeping sickness pathology and guidance towards innovative treatments. The present review aims to comprehensively describe the T. brucei biology and identify targets for new or commercialized antitrypanosomal drugs. Recent metabolomic applications to provide a deeper knowledge about the mechanisms of action of drugs or potential drugs against T. brucei are highlighted. Additionally, the advantages of metabolomics, alone or combined with other methods, are discussed. Compared to other parasites, only few studies employing metabolomics have to date been reported on Trypanosoma brucei. Published metabolic studies, treatments and modes of action are discussed. The main interest is to evaluate the metabolomics contribution to the understanding of T. brucei's metabolism.

Single cell and spatial transcriptomic analyses reveal microglia-plasma cell crosstalk in the brain during Trypanosoma brucei infection.

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Human African trypanosomiasis, or sleeping sickness, is caused by the protozoan parasite Trypanosoma brucei and induces profound reactivity of glial cells and neuroinflammation when the parasites colonise the central nervous system. However, the transcriptional and functional responses of the brain to chronic T. brucei infection remain poorly understood. By integrating single cell and spatial transcriptomics of the mouse brain, we identify that glial responses triggered by infection are readily detected in the proximity to the circumventricular organs, including the lateral and 3rd ventricle. This coincides with the spatial localisation of both slender and stumpy forms of T. brucei. Furthermore, in silico predictions and functional validations led us to identify a previously unknown crosstalk between homeostatic microglia and Cd138+ plasma cells mediated by IL-10 and B cell activating factor (BAFF) signalling. This study provides important insights and resources to improve understanding of the molecular and cellular responses in the brain during infection with African trypanosomes.

Association between IL1 gene polymorphism and human African trypanosomiasis in populations of sleeping sickness foci of southern Cameroon.

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Human African Trypanosomiasis (HAT) is a neglected tropical disease caused by infections due to Trypanosoma brucei subspecies. In addition to the well-established environmental and behavioural risks of becoming infected, there is evidence for a genetic component to the response to trypanosome infection. We undertook a candidate gene case-control study to investigate genetic associations further. We genotyped one polymorphism in each of seven genes (IL1A, IL1RN, IL4RN, IL6, HP, HPR, and HLA-G) in 73 cases and 250 controls collected from 19 ethno-linguistic subgroups stratified into three major ethno-linguistic groups, 2 pooled ethno-linguistic groups and 11 ethno-linguistic subgroups from three Cameroonian HAT foci. The seven polymorphic loci tested consisted of three SNPs, three variable numbers of tandem repeat (VNTR) and one INDEL. We found that the genotype (TT) and minor allele (T) of IL1A gene as well as the genotype 1A3A of IL1RN were associated with an increased risk of getting Trypanosoma brucei gambiense and develop HAT when all data were analysed together and also when stratified by the three major ethno-linguistic groups, 2 pooled ethno-linguistic subgroups and 11 ethno-linguistic subgroups. This study revealed that one SNP rs1800794 of IL1A and one VNTR rs2234663 of IL1RN were associated with the increased risk to be infected by Trypanosoma brucei gambiense and develop sleeping sickness in southern Cameroon. The minor allele T and the genotype TT of SNP rs1800794 in IL1A as well as the genotype 1A3A of IL1RN rs2234663 VNTR seem to increase the risk of getting Trypanosoma brucei gambiense infections and develop sleeping sickness in southern Cameroon.

Trypanosoma brucei infection in a HIV positive Ugandan male.

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Human African Trypanosomiasis, or African Sleeping Sickness, is a parasitic infection caused by Trypanosoma brucei (gambiense or rhodesiense), and one of the declared neglected tropical diseases. Sleeping sickness has high fatality rates and is a continued threat in several African countries. We present characteristic clinical and microbiologic features of a fatal case of African Sleeping Sickness in an HIV-infected individual.